PROPOSAL FOR
Automated Distributor Sales Reporting & Pharmaceutical ERP Platform

ERP PLATFORM

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1.Executive Summary

The pharmaceutical industry relies heavily on accurate, timely and centralized sales information. Currently, the organization receives sales-related PDF reports from more than **48 distributors**, with each distributor potentially using a different document format, structure and terminology.

The absence of an automated mechanism to process these documents creates several challenges, including:

- Manual data entry and processing
- Time-consuming report preparation
- Risk of human error
- Difficulty consolidating distributor data
- Delayed management reporting
- Difficulty comparing distributor performance
- Lack of centralized sales intelligence
- Dependency on multiple existing software systems

We propose developing an **Automated Distributor Sales Reporting & Sales Intelligence Platform** that will allow authorized users to upload distributor PDF reports and automatically extract, standardize, validate and consolidate the information.

The platform will generate **SSR (Sales Stock Report, as per the organization's existing format)** along with multiple management and operational reports.
More importantly, the proposed solution will not be developed as an isolated reporting application. Its architecture will establish a **centralized business-data platform** that can progressively evolve into a complete **Pharmaceutical ERP System**, integrating Sales, Inventory, Procurement, Finance, HR, Distribution and other business operations.

Core vision

Solve the immediate sales-reporting problem today while building the foundation for the company's future ERP.

2.Current Business Problem

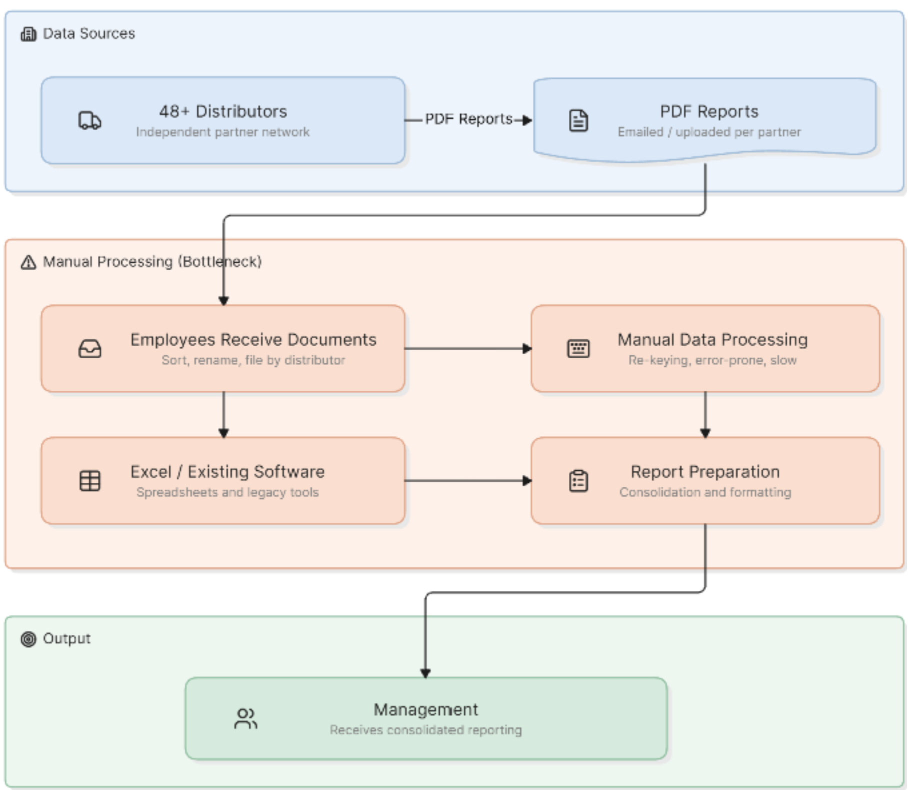
The organization currently receives sales information from more than 48 distributors in PDF format.

These documents may differ in:

- Layout
- Column names
- Product naming
- Product codes
- Date formats
- Quantity formats
- Pricing information
- Distributor-specific terminology
- Table structures

The current process requires employees to manually review and process these documents before meaningful reports can be generated.

Current workflow



This process becomes increasingly difficult as the number of distributors, products and transactions increases.

3. Key Challenges

3.1 Manual Processing

Employees spend significant time reviewing and entering information from distributor reports.

3.2 Multiple PDF Formats

Different distributors may provide completely different report formats.

3.3 Data Inconsistency

The same product may appear under different names or codes across distributors.

Example:

```
PANADOL TAB 500MG
PANADOL 500 MG TABLET
PANADOL TAB 500
```

All may represent the same product in the company’s master product catalog.

3.4 Reporting Delays

Management reports cannot be generated immediately because data must first be collected and processed.

3.5 Human Error

Manual data entry introduces the possibility of:

- Incorrect quantities
- Incorrect product mapping
- Duplicate records
- Incorrect totals
- Missing records

3.6 Lack of Centralized Data

Sales information is distributed across documents and existing systems instead of being available through one centralized platform.

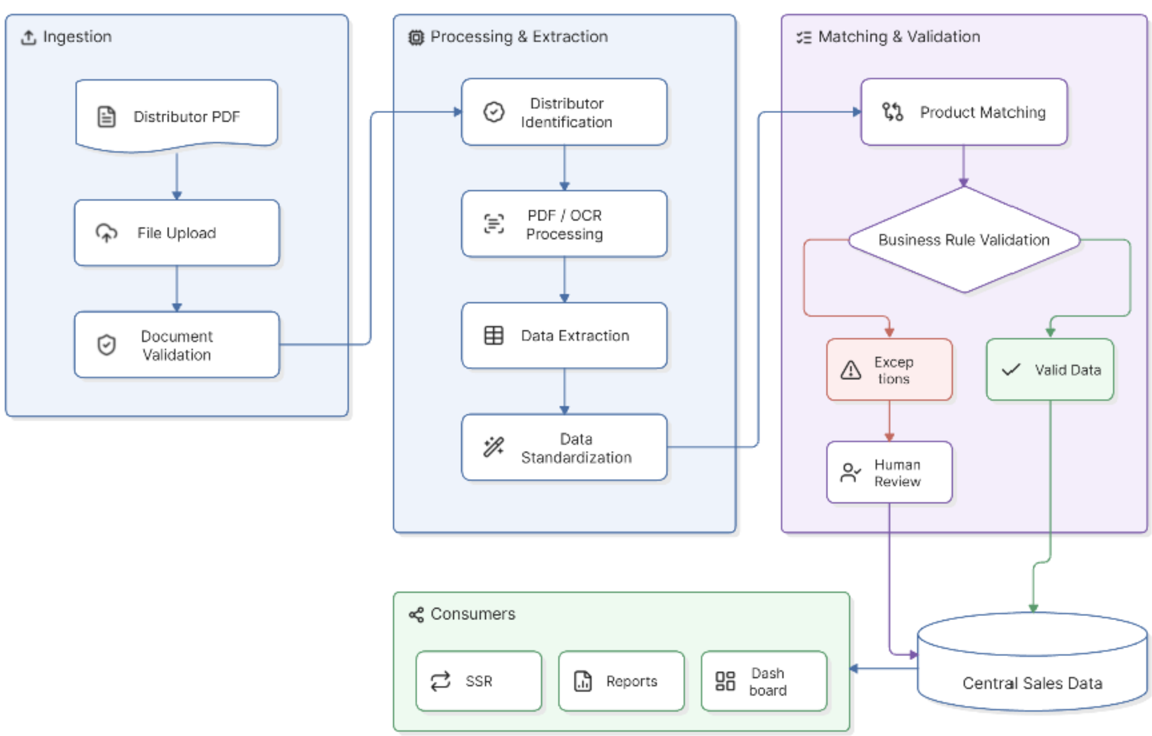
4. Proposed Solution

We propose developing a **Distributor Sales Intelligence Platform** with an integrated **Document Intelligence Engine**.

The system will allow authorized users to:

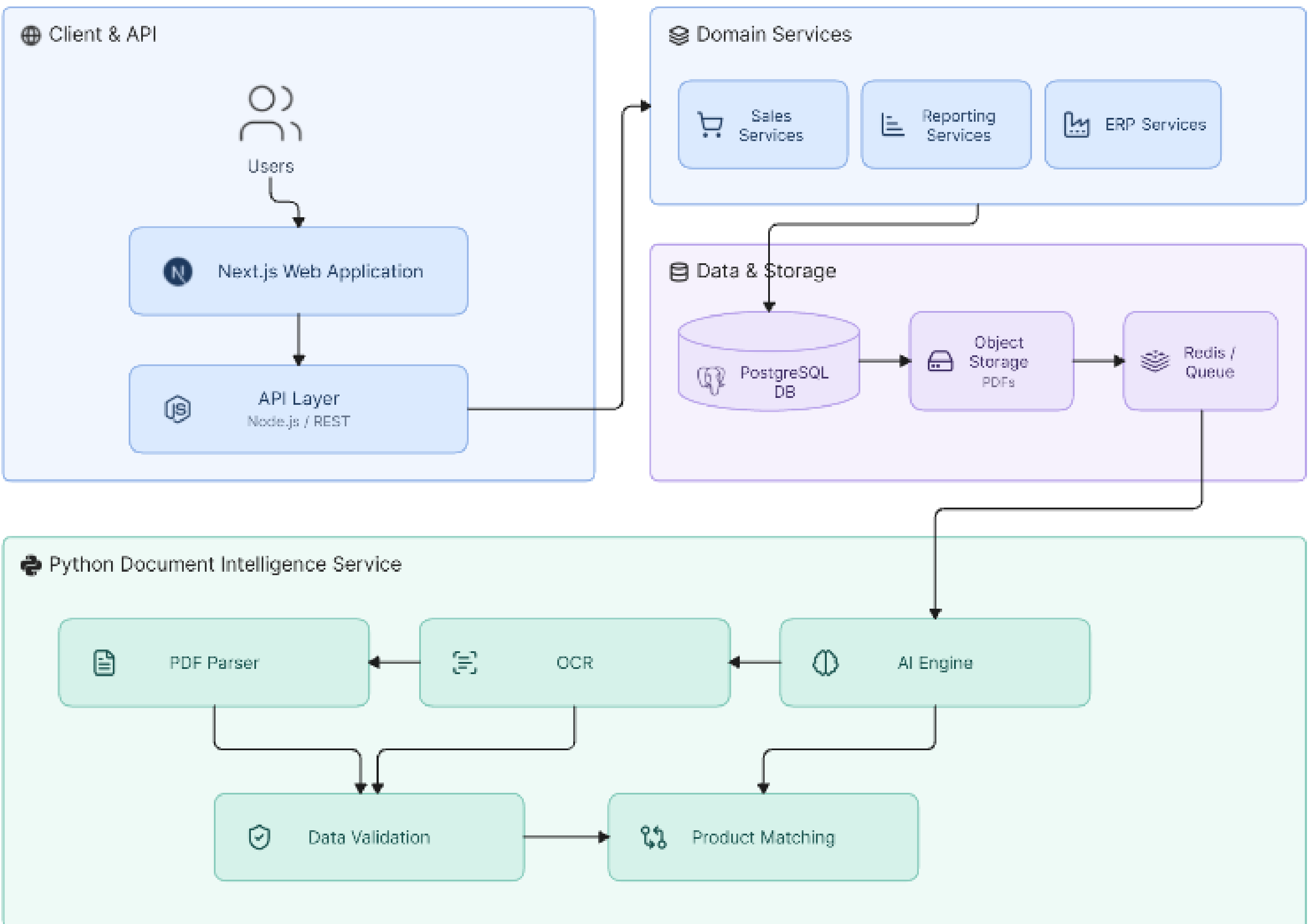
1. Upload distributor PDF reports
2. Automatically identify the distributor
3. Extract relevant information
4. Process scanned PDFs using OCR
5. Identify products
6. Map distributor products to the company's master product catalog
7. Validate extracted information
8. Flag exceptions for human review
9. Store standardized sales information
10. Generate SSR reports
11. Generate management dashboards
12. Export reports to Excel/PDF
13. Track document processing history
14. Maintain complete audit logs

5. Proposed Business Workflow



6. System Architecture

The proposed system will use a modular architecture designed to support both the initial sales-reporting requirements and future ERP modules.



7. Document Intelligence & PDF/AI Workflow

The core differentiating component of the platform will be the **Document Intelligence Engine**.

It will combine traditional document processing, OCR, business rules and AI-assisted data matching.

Step 1 — Upload

Users upload one or multiple PDF documents.

The system will validate:

- File type
- File size
- Document integrity
- Duplicate uploads
- User permissions

Step 2 — Distributor Identification

The system will identify the distributor using:

- Distributor metadata
- Document content
- Known document templates
- Distributor-specific identifiers
- Configured document rules

Step 3 — PDF Extraction

For digitally generated PDFs, the system will extract text and tables directly.

Potential technologies:

- PyMuPDF
- pdfplumber
- Camelot
- Tabula

Step 4 — OCR Processing

If a document is scanned or contains images, OCR processing will be triggered.

Possible technologies:

- Tesseract OCR
- Azure Document Intelligence
- AWS Textract
- Google Document AI

The final implementation will depend on accuracy requirements, infrastructure and data-security policies.

Step 5 — Data Standardization

Distributor-specific fields will be converted into the company’s standardized schema.

8. Product Matching Engine

A centralized **Product Master** will be maintained.

Example:

```
Product ID:        10234
Product Name:      Panadol
Strength:          500mg
Dosage Form:       Tablet
Pack Size:         20
Manufacturer:      XYZ
```

Distributor data may contain:

```
PANADOL TAB 500MG
PANADOL 500 MG TAB
PANADOL TABLET 500
```

The system will identify the appropriate master product using:

- Exact matching
- Product codes
- Fuzzy matching
- Configured mappings
- AI-assisted matching

Uncertain matches will be sent for human review.

9. Human Validation

Automation should not mean removing human control.

The system will provide a validation screen for exceptions.

Example:

Extracted Value	System Match	Status
PANADOL TAB 500	Panadol 500mg	✓
BRUFEN TAB	Brufen Tablet	✓
ABC-123	Unknown Product	⚠
XYZ-500	Possible Product	⚠

Users can correct the information and save the mapping.
The system can then reuse the mapping when similar documents are received in the future.

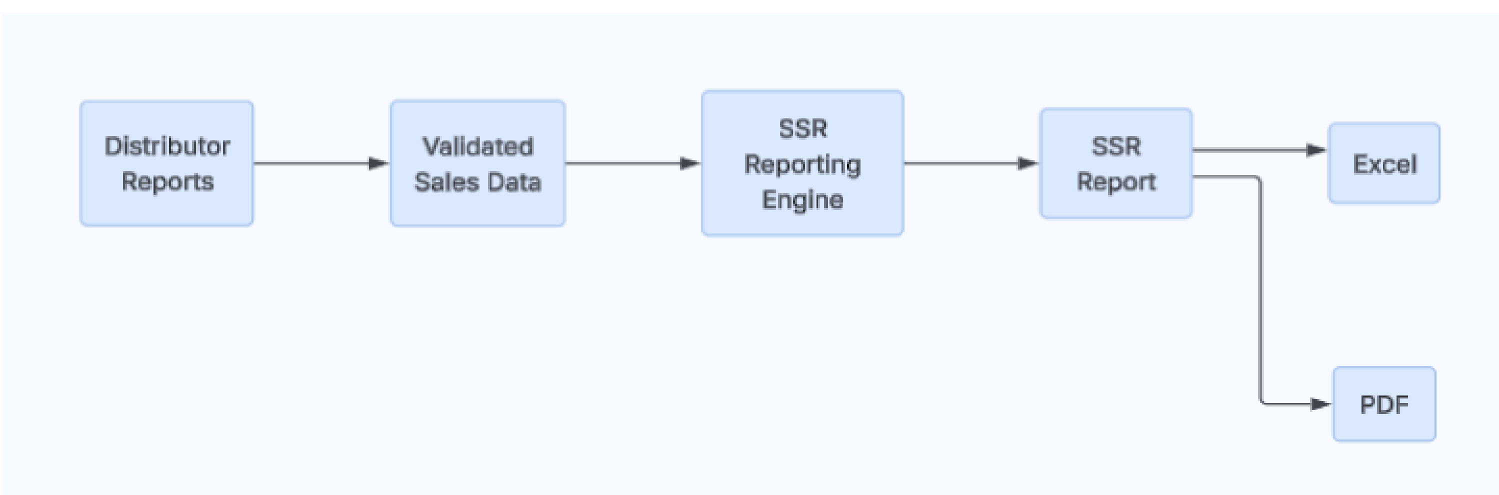
10. SSR Reporting

The platform will generate the organization's required **SSR report format** based on the business rules and fields confirmed during requirements analysis.

Possible SSR information may include:

- Distributor
- Reporting period
- Product
- Product code
- Quantity
- Sales value
- Returns
- Net sales
- Territory
- Region
- Sales representative
- Customer
- Other organization-specific fields

SSR workflow



11. Additional Reports

The same centralized sales data can be used to generate multiple reports.

Distributor-wise Sales

Compare sales performance across all distributors.

Product-wise Sales

Identify best-selling and slow-moving products.

Region-wise Sales

Analyze performance across geographical regions.

Territory-wise Sales

Evaluate territory performance.

Salesperson-wise Sales

Measure individual sales performance.

Monthly Sales

Track sales trends over time.

Target vs Achievement

Compare sales performance against defined targets.

Growth Analysis

Compare:

- Current month vs previous month
- Current year vs previous year
- Distributor growth
- Product growth

Returns Analysis

Analyze product returns and their impact on net sales.

12. Management Dashboard

A centralized dashboard will provide management with real-time visibility into sales performance.

Key Performance Indicators

```
Total Sales
Active Distributors
Products Sold
Monthly Sales
Growth %
Top Distributor
Top Product
Pending Documents
Processing Exceptions
```

Dashboard Filters

Users will be able to filter information by:

- Date
- Distributor
- Product
- Region
- Territory
- Salesperson
- Product category

13. Proposed Modules

Phase 1 — Sales Intelligence

User Management

- Users
- Roles
- Permissions
- Authentication
- Activity logs

Distributor Management

- Distributor master
- Distributor profiles
- Document formats
- Distributor mappings
- Distributor performance

Document Management

- PDF upload
- Bulk upload
- Document history
- Processing status
- Failed documents
- Document reprocessing

Document Intelligence

- PDF extraction
- OCR
- Data normalization
- Product matching
- Validation

Sales Management

- Sales records
- Sales imports
- Sales history
- Returns
- Adjustments

Reporting

- SSR
- Distributor reports
- Product reports
- Monthly sales
- Regional reports
- Salesperson reports
- Target vs achievement

Dashboard

- Management KPIs
- Charts
- Filters
- Drill-down reports

14. Future ERP Modules

The architecture will allow the following modules to be introduced progressively.

Sales & Distribution

- Customer management
- Sales orders
- Invoices
- Returns
- Sales targets
- Sales commissions
- Territory management

Inventory & Warehouse

- Warehouse management
- Stock management
- Batch management
- Expiry tracking
- Stock transfers
- Stock adjustments
- Stock alerts

Procurement

- Suppliers
- Purchase requests
- Purchase orders
- Goods receiving
- Supplier invoices
- Purchase history

Finance

- Accounts receivable
- Accounts payable
- Payments
- Expenses
- Ledgers
- Customer balances

- Distributor balances
- Financial reports

HR

- Employee management
- Departments
- Attendance
- Leave management
- Payroll
- Performance

15. Pharmaceutical-Specific Capabilities

As the platform evolves into a pharmaceutical ERP, the system can support industry-specific requirements such as:

- Batch tracking
- Expiry-date management
- Product recalls
- Warehouse stock
- FEFO-based inventory handling
- Returns management
- Damaged stock
- Near-expiry alerts
- Product movement history
- Distributor management
- Territory management
- Sales representative management

These capabilities will be designed according to the company's actual operational and regulatory requirements.

16. Technology Stack

Frontend

Next.js + TypeScript

- Modern enterprise web application
- Responsive interface
- High performance
- Scalable frontend architecture

UI

- Material UI
- Tailwind CSS
- Reusable design system

Backend

Node.js

Used for:

- Authentication
- Business APIs
- Sales management
- Distributor management
- Reporting
- ERP modules
- Integrations

Python + FastAPI

Used for:

- PDF processing
- OCR
- Document intelligence
- AI processing
- Product matching
- Data normalization

Database

PostgreSQL

Recommended for the ERP core because of:

- Strong relational capabilities
- Transaction integrity
- Reporting
- Complex relationships
- Scalability

Supporting Technologies

Requirement	Technology
Frontend	Next.js + TypeScript
UI	MUI + Tailwind
Backend	Node.js
AI/Document API	Python + FastAPI
Database	PostgreSQL
Cache	Redis
Background Jobs	BullMQ / Celery
PDF Processing	PyMuPDF / pdfplumber
Table Extraction	Camelot / Tabula
OCR	Tesseract / Cloud OCR
AI	LLM-based processing
File Storage	S3 / MinIO
API	REST / OpenAPI
Containers	Docker
Reverse Proxy	Nginx
CI/CD	GitLab CI/CD
Monitoring	Prometheus / Grafana
Version Control	Git

17. Security

Security will be considered a core requirement rather than an optional feature.

The platform will include:

- Role-based access control
- Secure authentication
- Password hashing
- HTTPS
- API authentication
- Input validation
- File validation
- Audit logging
- Database backups
- Access controls
- Secure document storage
- User activity tracking

Sensitive business information will only be accessible according to the user's assigned permissions.

18. Audit Trail

The system will maintain an audit trail for important activities.

Example:

```
User:          Admin
Action:        Approved Sales Report
Distributor:   ABC Distributor
Document:      July Sales Report
Date/Time:     01-Aug-2026 10:32 AM
```

This provides accountability and makes the platform suitable for enterprise operations.

19. Integration Strategy

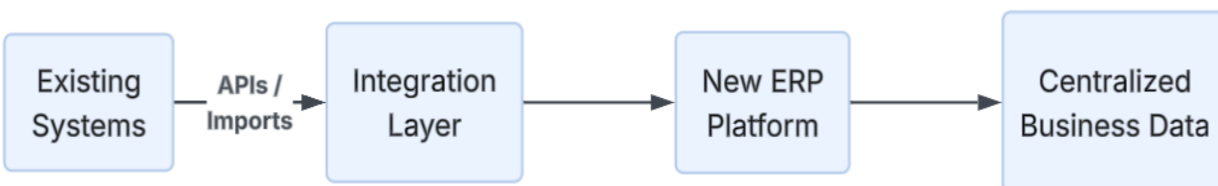
The organization already uses multiple software systems.

We therefore recommend an **integration-first approach** rather than immediately replacing all existing systems.

The new platform can integrate through:

- REST APIs
- Scheduled data synchronization
- Excel/CSV imports
- Database integration where appropriate
- Future API Gateway

Proposed approach



Over time, individual legacy systems can be replaced by ERP modules where appropriate.

20. Implementation Phases

Phase 0 — Discovery & POC

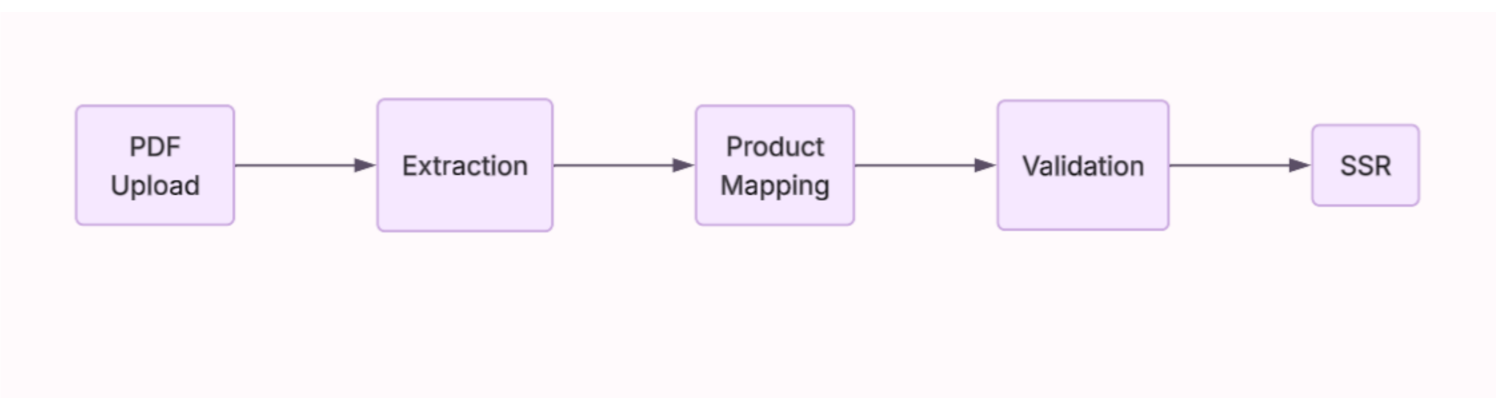
Duration: 1–2 weeks

Activities:

- Business process analysis
- Collect sample PDFs
- Analyze 5–10 distributors
- Understand current SSR
- Identify required fields
- Analyze existing systems
- Define integration requirements
- Develop technical architecture

Deliverable

Working proof of concept demonstrating:



21. Phase 1 — MVP Sales Intelligence Platform

Modules:

- Authentication
- User management
- Distributor management
- Product master
- PDF upload
- Document processing
- OCR
- Data extraction
- Product matching
- Validation
- Sales database
- SSR
- Basic dashboards
- Excel/PDF exports

Deliverable

A production-ready initial sales reporting platform.

22. Phase 2 — Advanced Sales Platform

Modules:

- Advanced dashboards
- Distributor analytics
- Product analytics
- Territory reports
- Salesperson reports
- Target vs achievement
- Returns
- Advanced filtering
- Scheduled reports
- Notifications
- Additional distributor templates
- API integrations

23. Phase 3 — Inventory & Distribution

Modules:

- Warehouse
- Inventory
- Batch management
- Expiry management
- Stock transfers
- Purchase orders
- Goods receiving
- Returns
- Stock alerts
- Distribution management

24. Phase 4 — Finance & Procurement

Modules:

- Suppliers
- Procurement
- Accounts payable
- Accounts receivable
- Payments
- Expenses
- Ledgers
- Financial reports
- Profitability analysis

25. Phase 5 — Complete ERP

Modules:

- HR
- Attendance
- Leave
- Payroll
- Advanced finance
- Advanced reporting
- Business intelligence
- Advanced integrations
- Mobile applications where required

26. Indicative Timeline

Phase	Duration
Discovery & POC	1–2 weeks
Phase 1 – Sales Intelligence MVP	6–8 weeks
Phase 2 – Advanced Sales	6–8 weeks
Phase 3 – Inventory	8–12 weeks
Phase 4 – Finance & Procurement	8–12 weeks
Phase 5 – Full ERP	10–16+ weeks

Important

The phases can be developed independently and deployed progressively.

The client does **not** need to wait for the complete ERP before receiving value from the system.

27. Recommended Commercial Strategy

Rather than asking the client to approve a large ERP project immediately, we recommend a phased approach.

Stage 1

Paid Discovery + POC

Demonstrate the PDF-to-SSR capability.

Stage 2

Sales Intelligence MVP

Deploy the first production system.

Stage 3

Advanced Sales & Distribution

Expand the system based on real operational usage.

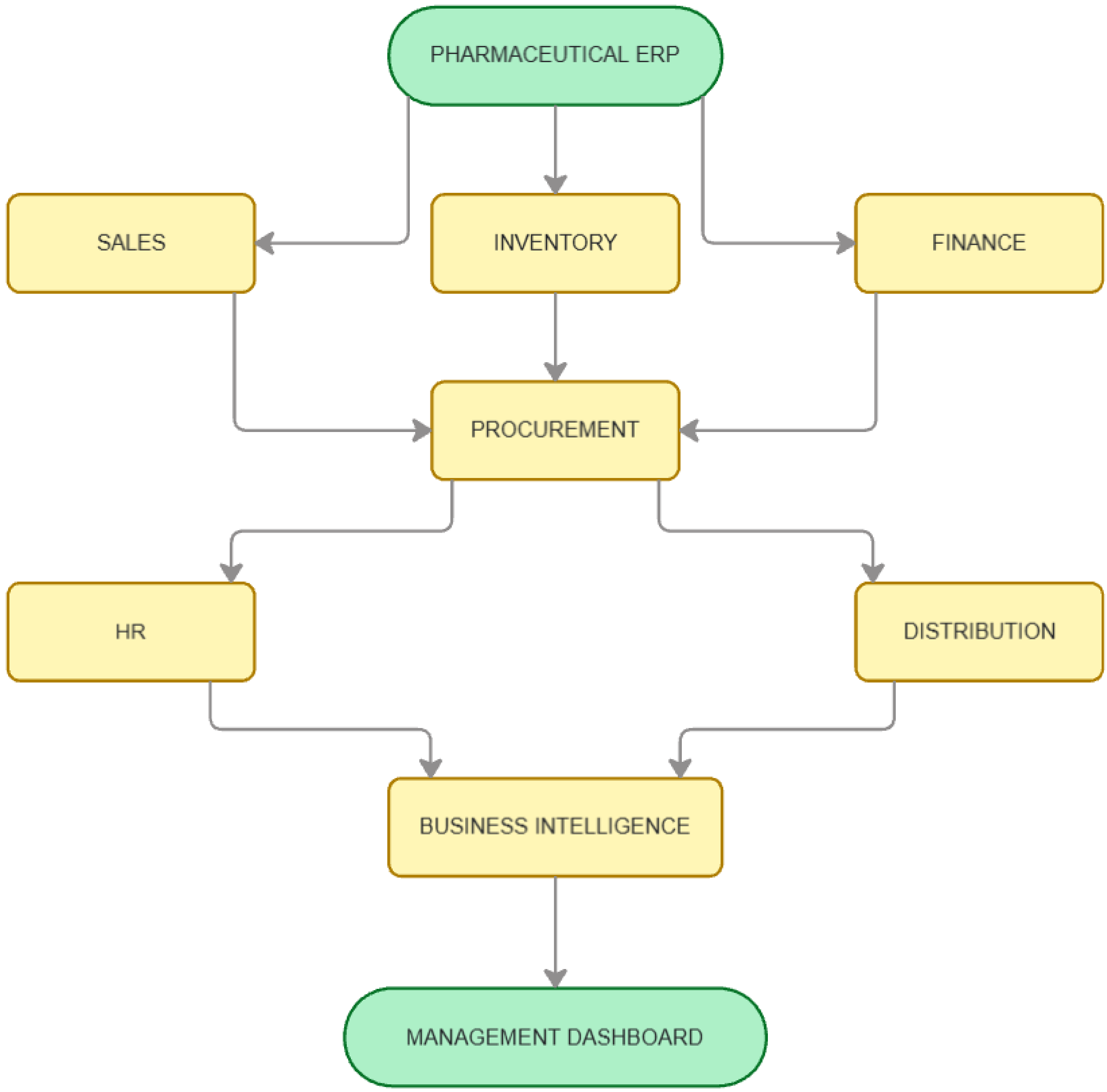
Stage 4

ERP Transformation

Gradually introduce Inventory, Procurement, Finance and HR.
This reduces the client's initial risk while creating a long-term technology partnership.

38. Future ERP Roadmap

The long-term vision is to transform the platform into a centralized enterprise management system.



The platform will eventually provide management with a single source of truth for the organization's operations.

29. Long-Term Business Intelligence

Once sufficient historical data is collected, the organization can move beyond traditional reporting toward predictive analytics.

Potential future capabilities include:

Sales Forecasting

Predict expected sales based on historical trends.

Product Demand Forecasting

Identify products likely to experience increased demand.

Inventory Optimization

Identify:

- Overstock
- Understock
- Slow-moving products
- Fast-moving products
- Near-expiry products

Distributor Performance Prediction

Identify distributors with declining performance.

Intelligent Alerts

Examples:

> "Distributor XYZ sales have decreased 18% compared with the previous month."
>
> "Product ABC has experienced a significant increase in demand."
>
> "Warehouse A contains products approaching expiry."

This transforms the platform from a reporting system into a **decision-support system**.

30. Key Benefits

The proposed platform will provide:

Operational Efficiency

Reduce manual processing of distributor reports.

Improved Accuracy

Automated extraction and validation reduce manual errors.

Faster Reporting

Management reports can be generated significantly faster.

Centralized Information

All sales information becomes available through one platform.

Better Decision Making

Management gets real-time visibility into sales performance.

Scalability

The platform can support additional distributors and business units.

Reduced Dependency on Legacy Systems

Existing systems can progressively be integrated and eventually replaced where appropriate.

Future ERP Foundation

The initial investment becomes the foundation for a complete enterprise management system.

31. Success Metrics

The project can be evaluated against measurable KPIs such as:

- Reduction in manual data-entry effort
- PDF processing time
- Data extraction accuracy
- Product matching accuracy
- Number of distributors automated
- Report generation time
- Number of manually corrected records
- System availability
- Management report turnaround time

The exact targets will be finalized during the discovery phase.

32. Recommended Next Step

We recommend beginning with a **Discovery & Proof-of-Concept phase**.

The client will provide representative sales reports from approximately **5–10 distributors**, including different PDF formats.
Our team will then:

1. Analyze the documents
2. Identify common data structures
3. Define the standard sales schema
4. Analyze the existing SSR format
5. Develop the document-processing prototype
6. Demonstrate product matching
7. Generate a sample SSR
8. Finalize the technical architecture
9. Prepare the detailed implementation plan
10. Submit the final development quotation

33. Conclusion

The proposed solution is not simply a PDF-uploading application.

It is the first step toward creating a **centralized digital platform for the organization's sales and business operations**.
The immediate objective is to automate distributor sales reporting and SSR generation.
The long-term objective is to build a scalable **Pharmaceutical ERP and Business Intelligence Platform** that connects:
Sales → Distribution → Inventory → Procurement → Finance → HR → Management
By implementing the solution in phases, the organization can achieve immediate operational benefits while progressively moving toward a unified and intelligent enterprise platform.

Proposed Vision

From Distributor PDFs to a Centralized Pharmaceutical ERP — Automated, Integrated and Data-Driven.

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Confidentiality:
This proposal contains preliminary technical and commercial information intended solely for the recipient organization. Final scope, pricing and implementation timelines will be confirmed following the discovery and proof-of-concept phase.